Tested algorithms

1) Naïve Spectral Subtraction Algorithm

$$|\hat{X}(\omega)| = \begin{cases} |Y(\omega)| - |\hat{D}(\omega)| & \text{if } |Y(\omega)| > |\hat{D}(\omega)| \\ 0 & \text{else} \end{cases}$$

- Where
  - X = filtered speech; Y = noisy speech; D = estimated noise
- Reconstruction (IFFT): Use the noisy signal phase (No difference if SNR>8dB)

Reference: P. C. Loizou. Speech Enhancement: Theory and Practice. CRCPress, 2013. ISBN 978-1-466-50421-9.

Tested algorithms

2) Spectral Over-subtraction Algorithm

$$|\hat{X}(\omega)|^2 = \begin{cases} |Y(\omega)|^2 - \alpha |\hat{D}(\omega)|^2 & \text{if } |Y(\omega)|^2 > (\alpha+\beta)|\hat{D}(\omega)|^2 \\ \beta |\hat{D}(\omega)|^2 & \text{else} \end{cases} \quad (5.41)$$

where
α is the oversubtraction factor ($\alpha \geq 1$)
β ($0 < \beta \ll 1$) is the spectral floor parameter

- Tries to minimize musical noise and non-linearities
- A and β should be function of the SNR for best results
- alpha=1.2, beta=0.002 used for simplicity

Reference: P. C. Loizou. Speech Enhancement: Theory and Practice. CRCPress, 2013. ISBN 978-1-466-50421-9.

Tested algorithms

3) Multiband Spectral Over-subtraction Algorithm

- Same as over-subtraction, but each frequency band has its alpha and beta

Reference: P. C. Loizou. Speech Enhancement: Theory and Practice. CRCPress, 2013. ISBN 978-1-466-50421-9.